

AudioBuddy: Demonstrating a Wearable and Mobile System Approach for Noise Sensitivity Awareness

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Abstract

This demo proposes a novel and practical application called AudioBuddy that uses sensing and tracking technology to facilitate awareness of physiological, psychological, and environmental indicators of potential noise sensitivity experiences. Existing research on noise sensitivity has proposed various approaches for supporting people's experiences, ranging from desensitization therapies to technological systems. However, there is a lack of systems that identify and track characteristics that may indicate the onset of noise sensitivity experiences so that people with noise sensitivity can take preventative actions before reaching stressed states and needing to self-regulate. AudioBuddy is a multi-device application that addresses this aim by leveraging the sensing capabilities of smartwatches for health monitoring, facilitating awareness and information sharing for community support, and presenting users with strategies for managing and regulating noise sensitivity experiences.

CCS Concepts

• **Human-centered computing** → **Empirical studies in accessibility**; **Accessibility technologies**.

Keywords

Sensors; Smartwatch; Health Monitoring; Noise Sensitivity

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1 Introduction

Noise sensitivity, described as one or a combination of hyperacusis, misophonia, tinnitus, and phonophobia¹, impacts many people globally, including many people with autism and ADHD [8, 12, 16]. Further, studies have shown relationships between noise sensitivity and various psychophysiological responses (e.g., changes in heart rate, skin conductance, affective states, etc.) due to environmental noise [14, 15]. Existing research and clinical practice around noise sensitivity have proposed various approaches for supporting people's experiences, ranging from desensitization therapies to technological systems. In our work, we focus instead on identifying and tracking characteristics that may indicate the onset of noise sensitivity experiences so that people with noise sensitivity (PWNS) can take preventative actions before reaching stressed states and needing to self-regulate. Awareness of the psychophysiological effects of environmental stimulants can inform when and how to support PWNS. Thus, we explore the potential of mobile and wearable systems to facilitate this awareness for PWNS and their care networks. In particular, wearable and mobile systems, such as mobile (e.g., [2, 3, 11]) and smartwatch applications (e.g., [4, 5, 13]), are regularly used to aid in tracking relevant personal data, which increases awareness and knowledge to inform behavior changes regarding an individual's health and well-being. Research in this area has been active for several years within the field of Personal Informatics (PI), yet few contributions of personal informatics artifacts have been made [9]. In this case, we designed and developed AudioBuddy (Figure 1) to propose a practical and novel PI application using commercially available devices to facilitate



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¹Hyperacusis is the perception of everyday sounds as excessively loud and/or painful; Misophonia is an aversion to specific sounds like mouth noises, repetitive sounds, etc.; Tinnitus is the perception of sound in the absence of sound/ringing in the ear; and Phonophobia is the fear of sound.

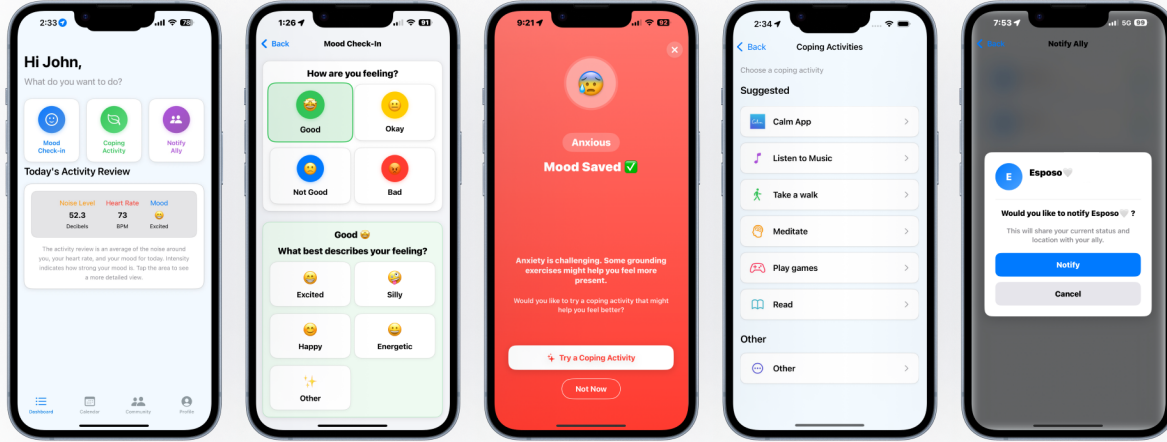


Figure 1: AudioBuddy System Screenshots. 1) Homescreen with navigation to Mood-Checkin, Coping Activities, Notify Ally, and an overview of the day’s Activity Review. 2) Mood Check-in. 3) Suggestions to do a coping activity. 4) Coping Activities. 5) Notify Ally

awareness among PWNS and their care networks of the potential effects of noise on their psychophysiological well-being.

2 AudioBuddy System

Through a multi-stage study consisting of an online forum analysis [7], interviews [8], and co-design workshops [10], we collaborated with PWNS and those in their care network to inform the design of AudioBuddy. The AudioBuddy system utilizes an Apple Watch to collect physiological (e.g., heart rate) and environmental data (e.g., noise level), and an iPhone that tracks self-reported psychological information and provides data visualizations. The app aims to raise awareness of characteristics that can indicate distress due to noise sensitivity and provide strategies for regulating when needed. The app’s features are explained below.

Health Monitoring features include heart rate tracking, environmental noise level monitoring, and real-time health data visualizations (Figure 1 Screen 1 and Figure 2 Screen 4). Prior work has indicated correlations between increased heart rate and noise sensitivity [6], as well as interest by PWNS to monitor their heart rate to understand how their body may respond during dysregulated states caused by noise [10]. Therefore, we focus on monitoring heart rate in this preliminary version of the app. In particular, the system integrates with Apple’s Health app and notifies end users when measurements exceed personalized noise and heart rate thresholds, which are set by the user within their profile (see Figure 2 Screen 3).

The Apple Watch captures heart rate data throughout the day, and measurements are published to the Health App every 5 minutes [1]. Additionally, environmental sound levels are continuously measured in 30-second intervals via the Noise App and are published to the Health App. AudioBuddy queries both heart rate and noise level data from the Health App in 1-minute and 30-minute intervals, respectively, incorporating this information into the user’s Activity Report for comprehensive health monitoring.

Personalized Activity Review and Reports include a daily snapshot summary of the data tracked by the system (Figure 2 Screen 4). This includes the heart rate and noise level trends shown on a time series graph, as well as a summary of their mood and coping activities for the day. Custom reports can be generated to view this data over a range of days, enabling users to gain insights on trends over a period of time (Figure 2 Screens 5 and 6).

Mood Check-ins encourage end users to reflect on their affective states, which can be impacted when exposed to intolerable sounds. Daily, the system sends notifications to do a mood check-in at 8 am, 12 pm, 4 pm, and 8 pm. This feature contains four moods: “Good,” “Okay,” “Not Good,” or “Bad,” which consist of four emotions that can be used to describe how end users are feeling. For example, if the user selects “Not Good” the four emotions that they can choose from are “Annoyed,” “Sick,” “Sad,” or “Tired.” If the four emotions provided are not descriptive of the person’s affective state, there is an “Other” option, and a different emotion can be typed.

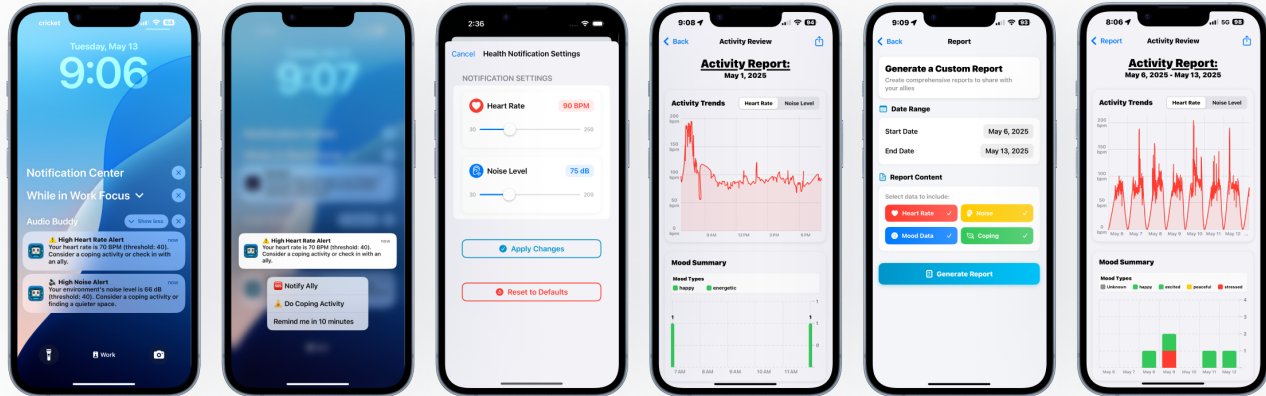


Figure 2: 1) Increased Heart Rate and Noise Level Notifications 2) Options for handling notifications 3) Personalized Heart Rate and Noise Level Threshold 4) Activity Report 5) Generate Custom Report 6) View of Generated Report

After selecting a mood, the intensity of that emotion is chosen on a scale from 1 to 10, with higher values indicating greater intensity. Finally, a journal entry with the question “*What’s on your mind?*” requires users to reflect and provide a more descriptive understanding of their affective state. When affective states that are “Not Good” or “Bad” are entered, the system suggests engaging in a coping activity (Figure 1 Screen 3). Once mood check-ins are completed, this data is then stored within the app and users can view their latest mood on the home screen under “Today’s Activity Review” (Figure 1 Screen 1) or a summary of their moods for the day on the “Activity Report” screen (Figure 1 Screen 4).

Coping Activities provide self-regulation options to engage in during stressful states (Figure 1 Screen 4). Coping activities are recommended when users enter a mood that is listed under the “Not Good,” or “Bad” categories, or when the app notifies the user due to an increase in their heart rate or the noise level that exceeds their thresholds (Figure 2 Screen 2). In any of these cases, if the user does not need or want to engage in a coping activity, they can choose to dismiss these suggestions. Users can also engage in coping activities without being prompted by the app. In this event, users can choose from a list of strategies, some of which open other apps. After selecting an activity, the user can engage in it, and the app will follow up after 15 minutes to determine whether the activity was effective in supporting their regulation needs through a mood check-in.

Community Support/Notify Ally enables users to contact their ally (e.g., parents/caregivers, spouses/partners, friends, family members, etc.) via the app through either a text message or email to receive help when self-regulation is difficult or to inform the ally

when they are in a stressful situation (Figure 1 Screen 5). These notifications provide the ally with the user’s location and latest available health data (e.g., heart rate and noise level) with further instructions for the ally to reach out to them.

3 Demo

We will demonstrate our system by guiding attendees through two primary use scenarios focused on supporting self and joint awareness of noise sensitivity experiences. In event-triggered scenarios, AudioBuddy, while tracking the user’s heart rate or the environmental noise level, sends a notification due to an increase in noise or the person’s heart rate (Figure 2 Screen 1). Attendees will set the threshold values to their desired levels (Figure 2 Screen 3) to trigger the notifications. Testing different thresholds will also give attendees an opportunity to consider the noise environment around them during the demo. These notifications will then give the attendees the options to either engage in a coping activity if needed, notify an ally, be reminded later, or dismiss the notification altogether (Figure 2 Screen 2). From these interactions, the attendees can then explore two other features of the app: Coping Activity and Notify Ally.

To demonstrate the real-time data tracking and visualization features by the system, we will have a variety of test personas with sample data, and using the mobile device, attendees can view the heart rate and environmental noise monitoring via real-time health data visualizations as well as the custom generated reports for each persona shown within the app (Figure 2 Screens 4-6).

In the second scenario, attendees can engage with the app independently (*i.e.*, without a notification). To demonstrate the system's ability to facilitate awareness of emotional well-being, attendees can use the mobile device to do a mood check-in and reflect on their affective states.

During the demonstration, we expect to engage with the attendees in discussions regarding the practical and everyday use of the system by people with noise sensitivity, potential limitations to our approach, alternative methods, and improvements in the health monitoring and data visualizations.

4 Conclusion and Future Work

Our demo shows the key features of the application and use cases for facilitating awareness of noise sensitivity. We are currently conducting user testing through a pilot deployment to understand the feasibility and acceptability of AudioBuddy to facilitate awareness of characteristics that may indicate the onset of noise sensitivity experiences and their impacts. While we evaluate our application, we will consider approaches to improve the system and achieve our research aims.

Acknowledgments

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